

ROROS, Norway

WMO: 012880

Lat: 62.577N

Lon: 11.352E

Elev: 2067

StdP: 13.63

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.0	(c) -18.7	(d) -30.6	(e) 1.1	(f) -25.0	(g) -23.6	(h) 1.6	(i) -18.6	(j) 28.9	(k) 25.9	(l) 25.7	(m) 25.7	(n) 0.9	(o) 320	(p) 0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.8	(c) 74.5	(d) 57.7	(e) 70.3	(f) 56.3	(g) 66.6	(h) 54.8	(i) 60.0	(j) 69.6	(k) 58.2	(l) 66.7	(m) 56.3	(n) 64.3	(o) 7.7	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.3	72.8	62.6	54.5	68.1	60.9	52.7	63.9	59.3	27.3	69.3	26.1	66.9	24.8	64.3	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.8	(b) 19.5	(c) 16.7	(e) -29.1	(f) 79.5	(g) 8.2	(h) 3.5	(i) -35.1	(j) 82.0	(k) -39.9	(l) 84.0	(m) -44.5	(n) 86.0	(o) -50.5	(p) 88.5		
(a) 22.8	(b) 19.5	(c) 16.7	(e) -28.5	(f) 62.7	(g) 8.5	(h) 2.2	(i) -34.6	(j) 64.3	(k) -39.6	(l) 65.6	(m) -44.4	(n) 66.9	(o) -50.6	(p) 68.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	13.7	16.9	22.5	32.8	41.3	48.5	53.8	51.4	44.6	34.1	24.6	17.0
	DBStd	16.63	14.16	12.01	11.02	6.85	7.15	6.34	5.62	5.20	5.46	7.18	10.19	13.72
	HDD50	6333	1127	927	851	516	286	102	25	45	176	493	763	1022
	HDD65	11490	1592	1347	1316	966	734	494	349	422	613	957	1213	1487
	CDD50	320	0	0	0	0	17	58	143	88	13	1	0	0
	CDD65	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH74	122	0	0	0	0	7	32	74	9	0	0	0	0
	CDH80	9	0	0	0	0	0	1	8	0	0	0	0	0
(14) Wind	WSAvg	6.8	6.8	6.7	7.5	7.0	7.7	7.7	6.6	6.1	6.7	6.4	5.9	6.3
(15) Precipitation	PrecAvg	24.4	1.9	1.6	1.4	1.1	1.8	2.6	3.3	3.3	2.1	1.9	1.7	1.7
	PrecMax	32.8	4.3	2.6	2.9	2.5	3.1	4.8	5.4	5.7	3.9	2.9	4.0	3.3
	PrecMin	19.3	0.2	0.4	0.4	0.5	0.6	1.1	1.2	1.1	0.8	0.6	0.3	0.7
	PrecStd	2.8	1.0	0.7	0.6	0.5	0.7	1.0	1.2	1.1	0.9	0.6	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	40.5	47.1	58.0	73.9	77.9	81.1	75.1	66.2	56.2	48.2	40.9
		MCWB	35.9	35.6	38.5	46.6	54.3	56.4	62.1	60.2	54.5	50.0	43.5	37.3
	2%	DB	36.5	37.1	42.4	53.1	66.8	73.0	75.6	70.0	61.1	51.1	43.4	38.6
		MCWB	33.9	33.6	36.8	43.0	52.2	56.0	59.1	56.7	52.9	45.7	40.2	35.6
	5%	DB	34.0	35.0	39.0	48.5	61.8	67.4	71.4	66.5	58.1	48.2	40.0	36.6
		MCWB	31.5	32.0	34.5	40.4	50.2	53.8	57.5	55.9	51.1	43.7	37.2	33.9
	10%	DB	31.4	32.5	36.6	44.5	55.9	62.7	67.0	62.9	55.2	45.4	36.9	34.0
		MCWB	29.5	30.0	32.8	38.1	46.6	51.7	55.8	54.5	49.7	41.9	34.5	31.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	36.6	39.5	47.2	56.9	59.4	64.0	61.0	56.4	51.8	44.2	37.8
		MCDB	38.2	39.9	44.4	56.7	68.3	69.3	75.5	69.4	62.4	54.7	48.1	40.0
	2%	WB	34.1	34.1	37.4	43.7	53.7	57.1	61.1	59.1	54.4	46.6	40.5	35.8
		MCDB	36.4	36.7	42.4	51.2	65.8	69.1	72.6	66.9	59.3	50.1	43.5	38.3
	5%	WB	31.8	32.0	35.1	41.0	50.8	55.2	59.1	57.2	52.1	44.2	37.3	33.9
		MCDB	33.8	34.8	38.6	47.7	60.0	66.0	68.2	64.1	56.8	47.6	39.6	36.2
	10%	WB	29.4	30.0	33.0	38.5	47.7	52.9	57.1	55.4	49.7	42.0	34.8	31.9
		MCDB	31.3	32.3	36.2	44.0	55.7	61.7	65.2	61.6	54.2	45.4	36.6	33.8
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.0	19.6	17.7	18.3	19.1	19.8	19.1	17.1	13.9	13.3	16.3
		MCDBR	13.0	14.7	17.9	24.2	30.9	30.8	29.6	26.5	23.3	18.7	14.6	16.5
		MCWBR	11.3	12.4	13.1	15.3	17.0	15.2	14.8	14.9	14.9	13.7	12.0	14.4
	5% WB	MCDBR	12.5	14.2	16.4	22.9	27.4	27.1	26.0	21.8	20.9	16.8	14.3	16.1
		MCWBR	11.0	12.0	12.2	14.8	15.7	13.7	13.5	13.0	14.2	12.8	12.0	14.3
(40) Clear-Sky Solar Irradiance	taub		0.200	0.256	0.288	0.318	0.332	0.328	0.340	0.329	0.300	0.282	0.212	0.176
	taud		2.042	2.215	2.212	2.230	2.397	2.473	2.452	2.472	2.542	2.442	2.102	1.954
	Ebn at Noon		154	225	259	274	281	284	277	271	260	218	145	95
	Edh at Noon		10	19	29	35	32	31	30	28	21	16	9	6
(44) All-Sky Solar Radiation	RadAvg		72	278	751	1342	1554	1554	1461	1128	709	316	97	37
	RadStd		8	33	63	87	89	134	153	109	68	40	14	3

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A

Nomenclature: See separate page